

Tutorial-IV (Mathematical Background)

1. Find the points on elliptic curve E_{23} (1.1)
2. Perform the addition on following 2D points

$$P = (3, 10) \quad Q = (9, 7)$$

3. Consider E_{13} (1. 1) and perform the following scalar multiplication and show the resultant point obtained.

(i) $4 * P$ where $P = (1, 4)$

(ii) $3 * Q$ where $Q = (8, 1)$